Exploring Collaboration between a Language and a Non-Language Agent

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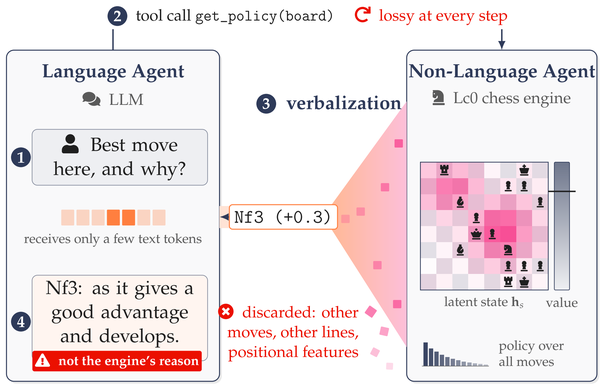
Motivation

How can an LLM effectively collaborate with non-language agents?

- LLMs have become general-purpose orchestrators that solve complex, domain-specific tasks by coordinating with tools and domain-expert subagents.
- However, in many important domains such as robotics and game playing, the strongest available agents are **non-language** neural networks (e.g., AlphaZero, RT-1).
- We study the collaboration between LLMs and non-language agents using **chess** as a controlled environment, owing to strong open engines such as Leela Chess Zero, public datasets, and real-world tasks.

Verbalization Is Lossy

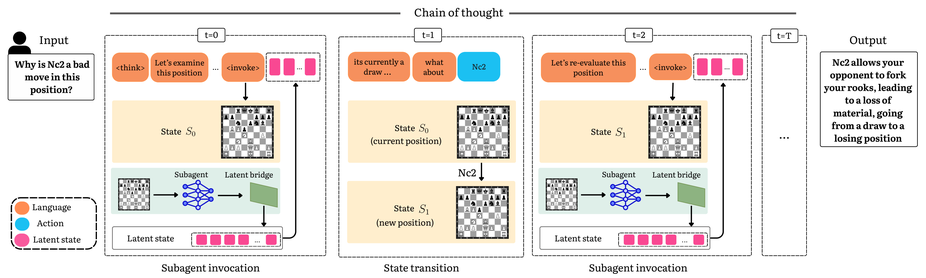
- Current methods call the agent as a tool and **verbalize** its output into text. However, this collapses the rich internal representation into a few text tokens.



Our Contributions

- **Verbalization Debt:** We propose the first quantification of the performance gap caused by verbalization.
- **LLAMIA-BENCH:** We introduce a dataset and benchmark spanning six chess tasks that rigorously test collaboration between LLMs and non-language agents.
- **LLAMIA:** We introduce a novel method to enable latent collaboration between pretrained LLMs and non-language agents. LLAMIA-14B outperforms verbalized baselines and existing state-of-the-art methods, including GPT-5, on all tasks.
- **Emergent collaboration strategies:** We also find that training with internalization enables novel multi-agent collaboration strategies.

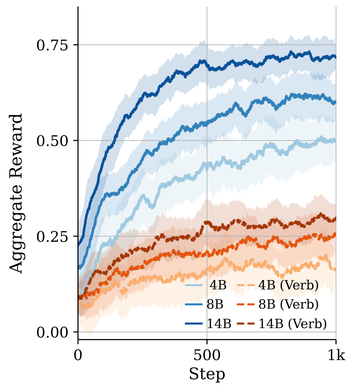
Latent State Internalization



- **Setup.** A **language agent** (Qwen3 LLM) collaborates with a frozen **non-language agent** (Lc0-BT4 chess engine); its internal state is projected into the LLM's chain of thought instead of being verbalized.
- **Projection.** Lc0's hidden state $\mathbf{h}_s$ (layer 14 of 15) is mapped by a 3-layer MLP, the **LatentBridge**, into the LLM's CoT at each invocation as k=32 **state tokens**: $\mathbf{z}_s = H_\varphi(\mathbf{h}_s)$.
- **Dynamic invocation.** Inside its **language** chain of thought, the LLM can repeatedly decide **when** to call the subagent and execute **actions** that update the board to counterfactual states, and call `get_policy` on any position. Each call re-encodes the current state, so the LLM always reasons over the position it is analyzing.

Training with Reinforcement Learning

- **Stage 1: projector alignment.** We follow LLaVA-style two-stage training, training only the LatentBridge on large-scale synthetic state–value pairs.
- **Stage 2: reinforcement learning.** We post-train the LLM and LatentBridge jointly with **DAPO** on six LLAMIA-BENCH tasks. **State tokens** are masked from the policy gradient. The engine stays frozen in both stages. The resulting model, **LLAMIA**, reasons over **language**, **action** and **state** tokens.
- **Verbalization Baseline:** We also train LLAMIA-VERB by replacing the state tokens with verbalized summaries (best moves, evaluation, position).



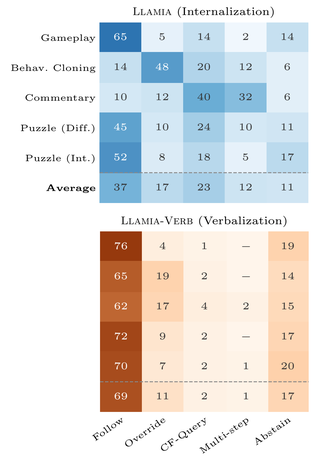
LLAMIA-BENCH

- We introduce **LLAMIA-BENCH**, a compilation of six popular chess tasks spanning Behavior Cloning, Move/Game Commentary Generation, and Puzzle Understanding, including three new datasets.
- None of these tasks can be solved by pretrained LLMs or engines alone.

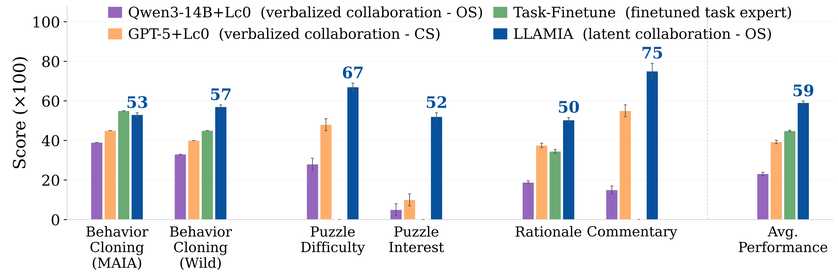
Behavior Cloning — Move a 1500-rated human would play: Nf3 (Elo 1500) — 12M games, 5 Elo bands	**Puzzle Difficulty** — Difficulty for human solvers: 1850 (+ solution) — 4M Lichess puzzles	**Move Rationale** — Why did the player make this move? (move Bb5) — 90K annotated games
BC in the Wild NEW — Same, for GMs, low clock, rating gaps: Qxd5 (+ context) — GM, low-time, Elo-gap games	**Puzzle Interest** NEW — How much players liked it: +72 votes (+ solution) — no text proxy exists	**Game Commentary** NEW — Commentate on this game (full game) — Agadmator-2K: 1.9K games

Emergent Collaboration Strategies

- We annotate LLM–subagent collaboration patterns using GPT-4o, e.g., *Follow*, where the LLM simply follows the engine.
- LLAMIA **adapts its strategy to the task**: engine-follow for gameplay, consult-then-override for behavior cloning, counterfactual queries for commentary.
- LLAMIA-VERB **collapses to engine-follow** on every task (62–76%).

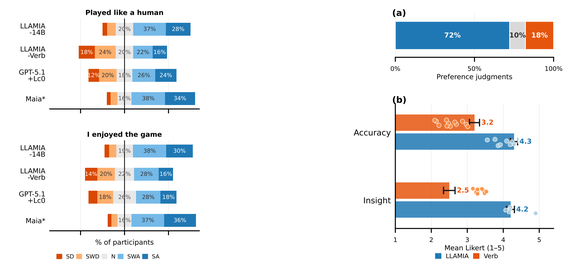


Results



- A single LLAMIA-14B model is best on all six tasks, ahead of GPT-5 with the same engine as a tool and even of state-of-the-art task-specific finetuned models.
- **Joint reasoning needs latent state.** On harder tasks that require reasoning jointly over human behavior and the position, such as Puzzle Interest, only LLAMIA scores well; verbalization catches up only on tasks a text summary can approximate, such as Puzzle Difficulty.
- **Beyond chess.** On Go with KataGo, the same method is ~10 points above the text-based control.

Human study



- **Human study** (n=12 players rated ≥1700): post-game ratings put LLAMIA on par with Maia* for human-likeness; 72% of commentary judgments prefer LLAMIA.